

Biology Unit 4: Cellular Respiration

Class notes, sample page for Memo Study Mode

Big picture

- Cellular respiration breaks down glucose to make ATP, the cell's energy currency.
- Overall equation: $\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O} + \text{energy (about 30–32 ATP per glucose)}$.
- It happens in three main stages: glycolysis, the Krebs cycle, and the electron transport chain.

1. Glycolysis

- Takes place in the cytoplasm and does not need oxygen.
- Splits one glucose (6 carbons) into two pyruvate molecules (3 carbons each).
- Net gain: 2 ATP and 2 NADH.

2. Krebs cycle (citric acid cycle)

- Takes place in the matrix of the mitochondria.
- Pyruvate is first converted to acetyl-CoA, releasing CO_2 .
- Each turn makes NADH and FADH_2 , which carry high-energy electrons to the next stage.
- The CO_2 we breathe out comes from this stage.

3. Electron transport chain

- Located in the inner mitochondrial membrane.
- Electrons from NADH and FADH_2 pass along protein complexes, pumping H^+ ions across the membrane.
- H^+ flows back through ATP synthase, which makes most of the ATP (about 26–28).
- Oxygen is the final electron acceptor and combines with H^+ to form water.

Without oxygen: fermentation

- Lactic acid fermentation happens in muscle cells during hard exercise.
- Alcoholic fermentation happens in yeast and produces ethanol and CO_2 .
- Fermentation only makes 2 ATP per glucose (from glycolysis), so it is much less efficient.